AMS

INDEX

Sl no	Description	Available (YES /NO)
1	Work order	✓
2	Geometrical test chart	✓
3	Axis Calibration Report	✓
4	Practical test report	H
5	Spindel stabilization report	
6	Spindle balancing report	
7	Hydraulic circuit diagram	
8	Pneumatic circuit diagram	
9	Minutes of meeting copy	✓
10	Machine checklist	✓

Quality Assurance Executive

Quality Assurance Chief

February 16, 2009

Minutes of Meeting held between M/s. Acutech., Pune & M/s. Ace Manufacturing Systems Ltd., Bangalore at AMS works on 16.02.2009

AMS
Mr. S. Madhukar
Mr. Nagaraj Bhat

ACUTECH
Mr. Paresh R.Deshmukh

Sub: Inspection of 1 No. CNC Vertical Machining Center Model MCV-400 (Machine No : M11-467)
Ref: Your purchase order no. AT/Pur/2009/07 Dated 11.02.2009

Mr. Paresh R.Deshmukh of M/s. Acutech visited AMS on 16th February'09 for the inspection of subject machine.

Following tests were conducted on the machine and the reports will be sent along with the machine.

1. AMS has done laser calibration of 2 axes.

2. Balbar test has conducted on X & Y axis

Geometrical tests, Heavy cut test, NASA test and rigid tapping tests will conducted by AMS and reports will be sent on 20.02.09.

Following are the highlights of the minutes:

1. Perpendicularity between table top surface and z-axis along YZ plane is in higher limit to be corrected before machine dispatch.
2. Machine to be dispatched only after getting Rotary table from M/s.UCAM and the same to be interfaced on the machine and properly tested.

AMS requests Acutech to arrange for the release of balance payment immediately.

AMS thanks Mr. Paresh R.Deshmukh for his kind co-operation extended during machine inspection.

For AMS

For ACUTECH

Work Order

Poornaltech

Customer	:	M/s. Acutech., Pune
Machine	:	CNC Vertical Machining Center Model MCV-400
P.O. No.	:	AT/Pur/2009/07 11.02.2009
O.A. No.	:	OA-11-431-09-00 13.02.2009
W.O. No.	:	WO-11-431-09-00 13.02.2009

Work Order

01.	Work Order No. & Date	:	WO-11-431-09-00 13.02.2009
	Order Acceptance No. & Date	:	OA-11-431-09-00 13.02.2009
02.	Customer Name	:	M/s. Acutech.,
03.	Address	:	S.No.49/1A, Opp. Shatrunjay Mandir, Katraj Kondhwa Road, Pune – 411 048.
04.	Phone No.	:	
05.	Fax No.	:	
06.	E-Mail	:	acutech_india@yahoo.co.in
07.	Excise Control Code No.	:	
08.	Sales Tax No. & Date	:	CST: TIN:
09.	Contact Persons	:	Mr.Paresh Deshmukh, Partner
10.	Our Quotation No. & Date	:	Offer dated 04.02.2009
11.	Our Tooled-up Proposal No. & Date	:	Not applicable
12.	Customer P.O. No. & Date	:	AT/Pur/2009/07 Dated 11.02.2009
13.	Machine	:	CNC Vertical Machining Center Model MCV-400
14.	Colour of Machine	:	Blue and light gray combination (Blue - Munsell No. 4PB4/10 Light Gray - Munsell No. 2.5y7/0.5)
15.	Scope of supply, Technical Specs & CNC features	:	As per Anneuxre-I
16.	Machine Inspection	:	Geometrical accuracy's and machine acceptance as per test chart & NASA Test.
17.	Component trials	:	Not applicable
18.	Price	:	Rs. 20,20,000/- (Rupees Twenty lakhs twenty thousand only)
19.	Delivery	:	Machine to be kept ready for inspection by 13.02.2009.
20.	Payment terms	:	30% advance along with the order. Balance payment along with 100% of taxes & duties to be paid against proforma invoice before machine dispatch.

...2

21.	Duties & Taxes	:	CED @ 10.3% (which includes 3% Education Cess) & CST @ 2% against Form-C. However, Taxes and Duties prevailing at the time of despatch will be applicable.
22.	Exchange variation	:	Not applicable
23.	Packing charges	:	Inclusive.
24.	Transportation	:	**Door delivery through any reliable carriers and freight to-pay basis.**
25.	Insurance	:	**To be arranged by Acutech. We will furnish despatch details.**
26.	Commissioning	:	Free of cost by our Engineers.
27.	Warranty	:	a) For Machine: 15 Months from the date of supply of the machine or 12 months from the date of installation at the customer end whichever is earlier.
		:	b) For Fanuc System & Drives: 27 months from the date of supply of machine OR 24 months from the date of installation at the customer end whichever is earlier.
28.	Training	:	Three persons for One week at our works. Training includes. b) Programming and Operation - 1 Person b) Electrical / Electronic / Mechanical trouble shooting and maintenance - 2 Persons

For **Ace Manufacturing Systems Ltd.,**

(P. Ramadas)
Managing Director

(L.S. Umesh)
Sr. Vice President

ANNEXURE - I
SPECIFICATION OF CNC VERTICAL MACHINING CENTER
MODEL MCV-400

Sl. No.	Description	Unit	MCV-400
1	**TABLE**		
1.1	TABLE SIZE	mm	900 x 450
1.2	T-SLOT DIM.	N x W x P	4 x 18 x 100
1.3	MAX. LOAD ON TABLE	Kgf	300
1.4	DIST. FROM FLOOR TO TABLE TOP	mm	925
2	**MACHINE CAPACITY**		
2.1	X - AXIS TRAVEL	mm	600
2.2	Y - AXIS TRAVEL	mm	430
2.3	Z - AXIS TRAVEL	mm	460
2.4	DIST. FROM SPINDLE FACE TO TABLE TOP	mm	160 - 620
2.5	DIST. FROM SPINDLE CENTER TO COLUMN GUIDEWAYS	mm	510
2.6	COOLANT SUPPLY-FLOOD COOLANT AROUND THE SPINDLE	lpm	**50**
3	**SPINDLE**		
3.1	SPINDLE NOSE TAPER		7 / 24 No.40
3.2	SPINDLE SPEED - DIRECT DRIVE	rpm	**80 - 8000**
3.3	SPINDLE BEARING DIA	mm	70
3.4	TOOL CLAMPING FORCE	Kgf	450
4	**FEED**		
4.1	RAPID TRAVERSE X, Y & Z AXES	m / min.	30 / 30 / 30
4.2	CUTTING FEED	mm / min.	1 - 10000
4.3	BALL SCREW DIA FOR X, Y & Z AXES	mm	40 / 40 / 40
4.4	GUIDEWAYS FOR X, Y & Z AXES		Linear Motion
5	**AUTOMATIC TOOL CHANGER**		
5.1	NUMBER OF TOOLS	Nos.	20
5.2	MAX. TOOL DIA	mm	80
5.3	MAX. TOOL DIA. WITH ADJACENT POCKETS EMPTY	mm	155
5.4	TOOL SELECTION		Bi - Directional
5.5	TOOL LENGTH MAX.	mm	250
5.6	TOOL WEIGHT MAX.	Kg.	8
5.7	CHIP TO CHIP TIME	Sec.	3.9 / 4.8
5.8	TOOL CHANGE SYSTEM		Twin Arm
5.9	TOOL SHANK TYPE		BT - 40
5.10	PULL STUD TYPE		DIN 69872 Form-B
6	**ACCURACY WITH ROTARY ENCODER**		
6.1	POSITIONING FOR X, Y & Z AXES AS PER VDI DGQ 3441	mm	0.015
6.2	POSITIONING FOR X, Y & Z AXES AS PER JIS B6330	mm	0.010
6.3	REPEATABILITY FOR X,Y&Z AXES AS PER VDI DGQ 3441	mm	± 0.005
6.4	REPEATABILITY FOR X, Y & Z AXES AS PER JIS B6330	mm	± 0.003
7	**MOTORS**		
7.1	SPINDLE POWER 30 MIN./CONT. (SIEMENS) - **AT BASE SPEED OF 2000 RPM**	kW	10.5 / 14.5
7.2	FEED MOTOR X, Y & Z AXES (SIEMENS)	Nm	14.4 / 14.4 / 14.4

ANNEXURE - I
SPECIFICATION OF CNC VERTICAL MACHINING CENTER
MODEL MCV-400

Sl. No.	Description	Unit	MCV-400
8	**INSTALLATION DATA - MACHINE**		
8.1	MACHINE WIDTH	mm	1900
8.2	MACHINE DEPTH	mm	3150
8.3	MACHINE HEIGHT	mm	3250
8.4	MACHINE WEIGHT - NET	Kgf	4100
8.5	POWER SUPPLY FOR BASIC MACHINE	Kva	14
8.6	CNC CONTROL (WITH 10.4" TFT COLOUR DISPLAY)		SIEMENS 802D SL PRO
8.7	PNEUMATIC PRESSURE REQUIRED	Kgf / Cm2	5
8.8	PNEUMATIC SUPPLY REQUIRED AT ATMOSPHERIC PR.	Lpm	300
9	**TANK CAPACITY**		
9.1	COOLANT TANK	litr	300
9.2	LUBRICATION TANK (USEFUL QUANTITY)	litr	5 (4)
9.3	HYDRAULIC POWER PACK	litr	-----
10	**MACHINE FEATURES / ACCESSORIES - STANDARD**		
10.1	PANEL COOLER FOR ELECTRICAL CABINET		
10.2	PNEUMATIC COUNTER BALANCING FOR Z - AXIS		
10.3	FAIL SAFE BRAKE FOR Z - AXIS		
10.4	FLEXIBLE COUPLING FOR X, Y & Z AXES		
10.5	STANDARD WORK LIGHT		
10.6	THREE TIER INDICATION LAMP TO INDICATE CYCLE RUNNING READY FOR NEXT CYCLE & MACHINE IN ALARM STATUS		
10.7	MANUAL PULSE GENERATOR		
10.8	RIGID TAPPING		
10.9	TOTAL MACHINE ENCLOSURE		
10.10	COOLANT TANK & CHIP TRAY		
10.11	CHIP DISPOSAL TO THE REAR SIDE OF THE MACHINE THROUGH TWO CHUTES ON THE SIDES		
10.12	MANUALS - ONE SET		
10.13	BALL SCREW STRETCHING FOR X & Y AXES		
11	**MACHINE FEATURES / ACCESSORIES - ADDITIONALY ORDERED BY M/S.ACUTECH., PUNE.**		
11.1	FLOOD COOLANT SYSTEM AROUND THE SPINDLE		
11.2	CHIP FLUSHING SYSTEM ON THE BASE		
11.3	COOLANT GUN		
11.4	4TH AXIS PROVISION WITH SIEMENS MOTOR & DRIVES + ACCESSORIES		

MACHINE HISTORY (SIEMENS SL)

Date:-

NOTE:- *CHECK THE MACHINE AS PER THE WORK ORDER*

MACHINE MODEL	MCV-400	
MACHINE NUMBER	M11-467.	
CUSTOMER NAME	M/s Acculech	
SYSTEM	SIEMENS SL	PRO / PLUS
MAIN OPTIONS	STDCLNT,C.CONV(SLAT/SCRAPER),D.CONV, CTS,HYFIX, PNFIX,AUTODOOR,ROT.TB,SRVATC,SRVMAG,APC,I.TABLE	

M/C BACKUP (PATH) - SERVER:

M/C BACKUP (PATH) : \\arya\E-Backup-II\ELECTRICAL MACHINE DATA\AMS DISPATCH MACHINE BACKUPS (2007-08)

SYSTEM	TYPE NUMBER	SERIAL NUMBER
SIEMEN SL	6FC5370-0AA00-~~2AA0~~ 3AA0	S 2VV1BX0736
FULL CNC KEYBOARD(NEW LAYOT)	6FC5303-0DM13-1AA0	S2VV0BV05621
SINUMERIC I/O MODULE PP	6FC5611-0CA01-0AA0	S2VV0BF050070
QTY : 1 , 2 , 3		
ACTIVE LINE MODULE 3PH 380-480V 50HZ	6SL3131-7TE21-6AA0	ST-V02005414 ASE00688508
DOUBLE MOTOR MODULE	6SL3121-2TE21-0AA0	ST-VN2084823
QTY : 1 , 2	X, Y, 2	ASE00810534
DOUBLE MOTOR MODULE	6SL3121-2TE21-8AA0	
QTY : 1 , 2	SPINDLE , X, Y, Z,	
SINGLE MOTOR MODULE 1X15/~~18~~ 5A	6SL3121-1TE15-0AA0	ST-V02031695
QTY : 1 , 2	ATC , 4TH AXIS	ASE00784294
SINGLE MOTOR MODULE	6SL3121-1TE23-0AA0	ST-V02028332
QTY : 1 , 2	SPINDLE MOTOR	ASE00689769
SINGLE MOTOR MODULE	6SL3121-1TE21-0AA0	ST-V92083088
QTY : 1 , 2	2 X	ASE00784468.
BASIC LINE FILTER	6SL3000-0BE21-6DA0	ASE00378865
HF REACTOR 16 KW	6SN1111-0AA00-0BA1	
ELECTRONIC HANDWHEEL(MPG)	6FC9320-5DC01	
SINAMICS SENSOR MODULE	6SL3055-0AA00-5BA1	
SPINDLE MOTOR		
AC MSD HOLLOW SHAFT SPI MOTOR	1PM6105-2LF84-1AR1	
COMPACT INDUCTION MOTOR	1PH7106-2QG02-0LJ0 → 1PH7107-2QG02-0LJ0	V04 "77780 09
AXES MOTORS		
SYNCHRONOUS SERVOMOTORS	1FK7083-5AF71-1FG2	1FK7083-5AF71-1FG2 V04777501 004 V04777501 001

MACHINE HISTORY (SIEMENS SL)

Date:-

NOTE:- *CHECK THE MACHINE AS PER THE WORK ORDER*

X Y		
SYNCHRONOUS SERVOMOTORS	1FK7083-5AF71-1FH0	V04777502002
Z		
SYNCHRONOUS SERVO MOTORS	1FK7042-5AF71-1FB2	
ATC , 4TH AXIS		
OTHERS		
SIGNAL CABLE PRE ASSEMBLED (FEED BACK)	6FX2002-1DC00-1AH0 QTY:	
FUER PROFIBUS BIS 12 MBIT/S	6ES7972-0BA41-0XA0 QTY:	
SIGNAL CABLE PRE ASSEMBLED IP20/IP67	6FX5002-2DC10-1AF0 QTY:	
SIGNAL CABLE PRE ASSEMBLED IP20/IP67	6FX5002-2DC10-1AH0 QTY:	
POWER CABLE PREASSEMBLED	6FX5002-5CA01-1AF0 QTY:	
POWER CABLE PREASSEMBLED (WT BRAKE)	6FX5002-5DA01-1AF0 QTY:	
POWER CONNECTOR	6SL3162-2MA00-0AA0 QTY:	
TRANSFORMER -1		
MAKE	TS	
PRIMARY VOLTAGE	415 V 3PHASE 50 HZ AC	
SECONDARY VOLTAGE	415 V 3 PHASE AC	
POWER IN KVA	12 / 14 KVA	
TRANSFORMER -2 (FOR REG POWER SUPPLY)		
MAKE	TS	
PRIMARY VOLTAGE	415 V 3PHASE 50 HZ AC	
SECONDARY VOLTAGE	200V 3PHASE DOUBLE TAP 50 HZ AC	
POWER	700VA	
REGULATED POWER SUPPLY		
MAKE	TELEMECANIQUE\OMRON	
INPUT VOLTAGE	200V 1 PHASE AC	
OUTPUT VOLTAGE	24 V DC	
OUTPUT CURRENT	10 AMPS	
TYPE OR MODEL	ABL 1RPM24100 \ S8VS 24024	
QUANTITY	2 NOS	
ISOLATION SWITCH		
MAKE	SALZER \ ABB \ TELEMEC	
TYPE	LB263 \ 243 \NS100	
RATED CURRENT	40A \ 63A\ 100A	
AIR COOLER		
MAKE	SUNBEAM APPLIENCES \ADVANCE	Accdesignons/6914
CONTROL VOLTAGE	230 V AC	
RATED CURRENT	2.6 AMPS \ 2.4A \ 2.8A\3.5A\3.8A	
SET TEMPERATURE	3 DEGREE	
POWER	1000 WATTS \ 500WATTS	1926 Kcal/hr
TYPE OR MODEL	CNC 4000E \ CNC 2000EG \ TURBO1000 R33\ TURBO 500 R50	IGL00
REFRIGERANT	R-134A	

MACHINE HISTORY (SIEMENS SL)

Date:-

NOTE:- *CHECK THE MACHINE AS PER THE WORK ORDER*

ITEM	TYPE NUMBER	SERIAL NUMBER
SURGE SUPPRESSORS		
MAKE	YSL	
TYPE OR MODEL	YSL-RC3-1 \ 022-575VAC	
QUANTITY	4 nos	
TERMINAL STATION		
MAKE	WAGO	TB1,TB3,TB5,TB7,TB9,TB10,TB11
		TB2,TB4,TB6,TB8,TB19
TOOL BREAKAGE SENSOR		
MAKE	OMRON TOUCH SWITCH	
TYPE OR MODEL	D5B 1515 / D5B1535	
CONTROL VOLTAGE	24 V DC(100ma MAX)	
QUANTITY		
D TYPE CONNECTOR (DUST COVERS)		
MAKE	OEN (DUST COVERS - ESSEN)	
9 PIN MALE (QTY)		
9 PIN FEMALE (QTY)	5 nos.	X20 X21 X22 X23 X24
		X25 X26 X21A X22A X29
25 PIN MALE (QTY)		I4 ,O1 KB
25 PIN FEMALE (QTY) [EMG MONITOR]	2 nos	EMG MONITOR,RS-232
37 PIN MALE (QTY)		I2 I5 I/O1
37 PIN FEMALE (QTY)		I3 I5A
50 PIN MALE (QTY)	4 nos	02 I1 O1(PCB SIDE),I1(MOP SID)
50 PIN FEMALE (QTY)	1 no.	O1(MOP SIDE)
HARTING CONNECTOR		
MAKE	SAINATH \ LAPP\ INDO ELECTRICAL	FROM CABINET (BACK SIDE)
COOLANT TANK	10 \ 16 \ 24 PIN	R.H.S : L.H.S:
HYDRAULIC POWER PACK	10 \ 16 \ 24 PIN	R.H.S : L.H.S:
CHIP CONVEYOR	10 \ 16 \ 24 PIN	R.H.S : L.H.S:
FUME EXTRACTOR	10 \ 16 \ 24 PIN	LOCATION:
OTHERS		
	10 \ 16 \ 24 PIN	LOCATION:
	10 \ 16 \ 24 PIN	LOCATION:

RELAYS / PCB	MAKE	MODEL	CRLVOL	QTY	PURPOSE / NO
SOLID STATE RELAY	ELEC.RELAY	SJDA - 4825 -00	24V DC	2 4 6	MAGZ ARM INDEX TBL
SOLID STATE RELAY (MODULE)	YSL	ACSSC-2P-1	24V DC	1	MAGZ
A.C.RELAY	OMRON	MK2P-I	230 AC 1P	1	PANEL COOLER
CNC ON/OFF PCB	YSL	RP-002-TC-CP-1	24V DC	1	CNC ON/OFF CONTROL
SINGLE RELAY MODULE	YSL	YSL-UMK-1RM DC-CN-FI-2	24V DC	1	
DOUBLE RELAY MODULE	YSL	YSL-UMK-2RM DCCN-FI-2	24V DC	2	STAR / DELTA INPUT AND OUTPUT
I/O PCB MACHINE	YSL	YSL-S26-009-013-REV1-12 / 04	24V DC	1	
I/O PCB MCP	YSL	YSL-S26-009-012-REV1-12 / 04	24V DC	1	
RELAYS / PCB	MAKE	MODEL	CRLVOL	QTY	PURPOSE / NO
SOLID STATE RELAY MODULE					
2-RELAY MODULE	YSL	YSL-S26-009-27-CN	24V DC		X21 X21A X26
4-RELAY MODULE	YSL	YSL-S26-009-26-CN	24V DC		X22 X26
8-RELAY MODULE	YSL	YSL-S26-009-28-CN	24V DC		X25 X26
RELAY BOARD WITH RELAYS	YSL	S26-009-23-CN	24 V DC		X20 X21 X22 X23 X24
POT. FREE	YSL	S26-010-14-CN	24 V DC		
LAMPS					
THREE TIER LAMP	PATANKAR / PATOLITE		24 V DC	LEDTYPE	
SINGLE TIER LAMP(GREEN)			24 V DC	LAMP TYPE	
MACHINE LAMP	PATHANKAR	DOUBLE LAMP	230 V AC	1 NO	
	PATHANKAR	SINGLE LAMP	230 V AC	2 Nos	
CABINET LAMP	PASSOLITE/ PHILIPS		230V AC	1 NO	
M.O.P ELEMENTS	MAKE	TYPE	QTY		PURPOSE
SEALED KEYBOARD \ MCP	YSL / SBL / FANUC		1		
APC MOP	YSL		1		
ROTARY SWITCH					
	TOSOKU	2POLE 4WAY (DPP03 003N20R)			MPG FEED SELECTION
	TOSOKU	4POLE 16WAY (DPP03 015J20R)			RAPID/FEED O/R
	TOSOKU	5POLE 10WAY (DPP03 009N20R)			MODE SELECTION
	OEN	5POLE 10WAY (A17402) AM3			MANUAL APC FUN SEL.

RELAYS / PCB	MAKE	MODEL	CRLVOL	QTY	PURPOSE / NO
SOLID STATE RELAY	ELEC.RELAY	SJDA - 4825 -00	24V DC	2 4 6	MAGZ ARM INDEX TBL
SOLID STATE RELAY (MODULE)	YSL	ACSSC-2P-1	24V DC	1	MAGZ
A.C.RELAY	OMRON	MK2P-I	230 AC 1P	1	PANEL COOLER
CNC ON/OFF PCB	YSL	RP-002-TC-CP-1	24V DC	1	CNC ON/OFF CONTROL
SINGLE RELAY MODULE	YSL	YSL-UMK-1RM DC-CN-FI-2	24V DC	1	
DOUBLE RELAY MODULE	YSL	YSL-UMK-2RM DCCN-FI-2	24V DC	2	STAR / DELTA INPUT AND OUTPUT
I/O PCB MACHINE	YSL	YSL-S26-009-013-REV1-12 / 04	24V DC	1	
I/O PCB MCP	YSL	YSL-S26-009-012-REV1-12 / 04	24V DC	1	
RELAYS / PCB	MAKE	MODEL	CRLVOL	QTY	PURPOSE / NO
SOLID STATE RELAY MODULE					
2-RELAY MODULE	YSL	YSL-S26-009-27-CN	24V DC		X21 X21A X26
4-RELAY MODULE	YSL	YSL-S26-009-26-CN	24V DC		X22 X26
8-RELAY MODULE	YSL	YSL-S26-009-28-CN	24V DC		X25 X26
RELAY BOARD WITH RELAYS	YSL	S26-009-23-CN	24 V DC	5 No	X20 X21 X22 X23 X24
POT. FREE	YSL	S26-010-14-CN	24 V DC		
LAMPS					
THREE TIER LAMP	PATANKAR / PATOLITE		24 V DC	LEDTYPE	
SINGLE TIER LAMP(GREEN)			24 V DC	LAMP TYPE	
MACHINE LAMP	PATHANKAR	DOUBLE LAMP	230 V AC	1 NO	
	PATHANKAR	SINGLE LAMP	230 V AC	2 Nos	
CABINET LAMP	PASSOLITE/ PHILIPS		230V AC	1 NO	
M.O.P ELEMENTS	MAKE	TYPE	QTY	PURPOSE	
SEALED KEYBOARD \ MCP	YSL / SBL / FANUC		1		
APC MOP	YSL		1		
ROTARY SWITCH					
	TOSOKU	2POLE 4WAY (DPP03 003N20R)	1 NO	MPG FEED SELECTION	
	TOSOKU	4POLE 16WAY (DPP03 015J20R)	1 NO	RAPID/FEED O/R	
	TOSOKU	5POLE 10WAY (DPP03 009N20R)	1 NO	MODE SELECTION	
	OEN	5POLE 10WAY (A17402) AM3		MANUAL APC FUN SEL.	

SELECTOR SWITCH				
	TEKNIC	WITH KEY		AUTO DOOR/ SER ARM/ MAG SEL / PRG SEL /SEAT CHECK / DNC
	TEKNIC	2 POSITION	1 no	DISPSEL CLNTGUN APC STN1 OR STN2
	TEKNIC	2 POSITION WITH SPRING RTN	—	CHIP CONV FWD \ REV
	ELCOM	EDIT KEY SWITCH (AB)		PROGRAM EDIT
	TEKNIC	TOGGLE SWITCH		
M.O.P ELEMENTS	MAKE	TYPE	QTY	PURPOSE
PUSH BUTTONS				
	TEKNIC	GREEN PUSH BUTTON	1 NO	CYC ON, CLP 1, CLP 2 A DOOR, APC ST EJCTR UP/DN, LOC FWD/REV
	TEKNIC	RED PUSH BUTTON	1 NO	CYC OFF,DCLP,APC STOP
	TEKNIC	BLACK PUSH BUTTON		
	TEKNIC	MUSHROOM PB(RED)	1 NO	MACHINE EMG, CCR EMG, DOUBLE HAND EMG,APC EMG
	TEKNIC	MUSHROOM PB(BLACK / GREEN)	—	DOUBLE HAND CYCLE START
	TEKNIC	LUMINIOUS PB(GREEN)	1 NO	TOOL DCLP / PAL SETUP (APC MOP)
ELEMENTS	MAKE	TYPE	QTY	PURPOSE
PROXIMITY SENSORS	Teknic	Yellow PB	1 no	Reset
	TEKNIC	M12 X 73 MM CONN TYPE	3 no's	A
	TEKNIC	M12 X 52 MM CONN TYPE		B
	TEKNIC	M12 X 40 MM NON CONN TYP		C
	TEKNIC	M12 X 68 MM NON CONN TYP		D
	TEKNIC	M8 X 50 MM NON CONN TYP		E
	YSL	M12 X 50 MM NON CONN TYP(PNP)		G
	TEKNIC	EGT12X04UP 024-2000L		F
	TEKNIC	M8 CONN TYPE		H
	TEKNIC	CABLE(SWLF3AP-5000P)		
	TEKNIC	CABLE(SWLF3AP-2000P)		
		SPINDLE ORIENTATION.PXS	(A) B C D E G F H	
		TOOL CLAMP PROXIMITY	(A) B C D E G F H	
		TOOL DECLAMP PROXIMITY	(A) B C D E G F H	
		ONE ROT.SIGNAL PROXIMITY	A B C D E G F H	
		MAG IN POSITION PROXIMITY	A B C D E G F H	
		NO TOOL IN SPINDLE PROXIMITY	A B C D E G F H	
		COOLANT LEVEL LOW	A B C D E G F H	
		CHIP CONVEYOR O/L PROXIMITY	A B C D E G F H	
		PROXIMITIES ON APC TABLE	A B C D E G F H	
		PROXIMITIES ON APC STATION	A B C D E G F H	
		CHIP CONVEYOR O/L PROXIMITY	A B C D E G F H	

ELEMENTS	MAKE	TYPE	QTY	PURPOSE
LIMIT SWITCH				
	TEKNIC\BCH	NG1RS-510 \ NLTPR	—	M/C DOOR CLOSE LIMIT SWITCH
	TEKNIC\BCH	NG1RS-510 \ NLTPR	—	CABINET DOOR CLOSE LIMIT SWT
	JAIBALAJI \ BCH	JLSSPR \ NLSPR	—	PALLET-1 STATION LIMIT SWITCH
	JAIBALAJI \ BCH	JLSSPR \ NLSPR	—	PALLET-2 STATION LIMIT SWITCH
	TEKNIC\BCH	NG1RS-510 \ NLTPR	—	CHIP CONV1 O/L LIMIT SWITCH
	TEKNIC\BCH	NG1RS-510 \ NLTPR	—	CHIP CONV2 O/L LIMIT SWITCH
	JAIBALAJI \ BCH	JLSSPR \ NLSPR	—	APC DOOR UP LIMIT SWITCH
	JAIBALAJI \ BCH	JLSSPR \ NLSPR	—	APC DOOR DOWN LIMIT SWITCH
	TELEMECHA	XCE-154 \ JAI BALAJI	—	CYCLE START LIMIT SWITCH
	TEKNIC	NIA K502	—	FIXTURE LIMIT SWITCH
	TEKNIC	SG-01 R552		FIXTURE LIMIT SWITCH
REED SWITCHES			—	
	FESTO / SMC	MACHINE DOOR CLOSE / FIXTURE		
SENSORS				
FLAT SENSOR	TEKNIC	FG2X05APO24-5000L	1	MAG REF POSITION / NO TOOL IN SPINDLE
FLOAT SENSOR	FILPRO	LI M 18 F	1 NO	COOLANT LEVEL LOW

ELEMENTS	MAKE	TYPE	QTY	PURPOSE
MCB'S				
		PROTEC / MERLIN GERIN / SIMENS		
F15		25A / 3P 40 / 3P 63A / 3P	1 NO	TRANSFORMER PRIMERY
F16		3A/3P	1 NO	REG. POWER SUPPLY
F17		30mA, 63A	—	RCCB FOR PWR CIRCUIT
F18		10A/1P	1 NO	230V AC SUPPLY
F19		40A/3P 63A/3P 100/3P	1 NO	TRANSFORMER SECONDARY
F20		1A/3P 3A/3P 5A/3P	1 NO	SPINDLE FAN / CP1A

ELEMENTS	MAKE	TYPE	QTY	PURPOSE
LIMIT SWITCH				
	TEKNIC\BCH	NG1RS-510 \ NLTPR		M/C DOOR CLOSE LIMIT SWITCH
	TEKNIC\BCH	NG1RS-510 \ NLTPR		CABINET DOOR CLOSE LIMIT SWT
	JAIBALAJI \ BCH	JLSSPR \ NLSPR		PALLET-1 STATION LIMIT SWITCH
	JAIBALAJI \ BCH	JLSSPR \ NLSPR		PALLET-2 STATION LIMIT SWITCH
	TEKNIC\BCH	NG1RS-510 \ NLTPR		CHIP CONV1 O/L LIMIT SWITCH
	TEKNIC\BCH	NG1RS-510 \ NLTPR		CHIP CONV2 O/L LIMIT SWITCH
	JAIBALAJI \ BCH	JLSSPR \ NLSPR		APC DOOR UP LIMIT SWITCH
	JAIBALAJI \ BCH	JLSSPR \ NLSPR		APC DOOR DOWN LIMIT SWITCH
	TELEMECHA	XCE-154 \ JAI BALAJI		CYCLE START LIMIT SWITCH
	TEKNIC	NIA K502		FIXTURE LIMIT SWITCH
	TEKNIC	SG-01 R552		FIXTURE LIMIT SWITCH
REED SWITCHES				
	FESTO / SMC	MACHINE DOOR CLOSE / FIXTURE		
SENSORS				
FLAT SENSOR	TEKNIC	FG2X05APO24-5000L	1	MAG REF POSITION / NO TOOL IN SPINDLE
FLOAT SENSOR	FILPRO	LI M 18 F		COOLANT LEVEL LOW

ELEMENTS	MAKE	TYPE	QTY	PURPOSE
MCB'S				
		PROTEC / MERLIN GERIN / SIMENS		
F15		25A / 3P 40 / 3P 63A / 3P		TRANSFORMER PRIMERY
F16		3A/3P		REG. POWER SUPPLY
F17		30mA, 63A		RCCB FOR PWR CIRCUIT
F18		10A/1P		230V AC SUPPLY
F19		40A/3P 63A/3P 100/3P		TRANSFORMER SECONDARY
F20		1A/3P 3A/3P 5A/3P		SPINDLE FAN / CP1A

MACHINE HISTORY

MAKE : SPRECHER & SCHUH / KLOCKNER MOELLER / SIEMENS

MPCB NO.	CONT. NO	MPCB RATING	MPCB SETTING	DESCRIPTION	MAKE	MODEL	RATING
F1	K1 & K2	1-1.6A	1.6A	MAGZINE MOTOR	PRAGATI	TORQUE MTR	0.9KVA
		2.5-4A	2.54A				1.9KVA
F2	K3	~~2.5-4A~~ 1.0-1.6A	~~2.5A~~ 1.6A	ARM MOTOR	PRAGATI	TORQUE MTR	1.6KVA
			3.2A				2.3KVA
F3	K4	0.63-1A	0.54A	LUB MTR	H.MTR MFG	INDUCTION MTR	0.09KW
		1A/3P	MCB				
F4	K5	1-1.6A	~~1.08A~~	FLOOD CLNT MTR	RAJMANE	RK3-270	0.37KW
		2.5-4A	3.4A		RAJMANE	RKM-206	1.5KW
					GRUNDFOS		2.25KW
F5	K6	1-1.6A	1.08A	CHIP PUSH CLNT MTR	RAJMANE	RK3-270	0.37KW
		2.5-4A	3.4A		RAJMANE	RKM-206	1.5KW
F6	K7	1-1.6A	~~1.08A~~	CHIP FLUSH CLNT MTR	RAJMANE	RK3-270	0.37KW
		2.5-4A	3.4A		RAJMANE	RKM-206	1.5KW
F7	K8	1-1.6A	1.08A	CHIP PUSH ON CHIP	RAJMANE	RK3-270	0.37KW
				CONVEYOR/CLNT GUN			
F8	K9	2.5-4A	3A	HYDRAULIC MTR	SIEMENS		1.1KW
		4-6.3A					
F9	K10	4-6.3A	6.18A	H.P.CLNT MTR	GRUNDFOS		2.2KW
		6-10A			B.BIJLEE		3.7KW
		10-16A			WANNER		

MACHINE HISTORY

MPCB NO.	CONT. NO	MPCB RATING	MPCB SETTING	DESCRIPTION	MAKE	MODEL	RATING
F10	K11 & K12	0.63-1A	0.78A	CHIP CONV.-1 MTR	LHPL / BONFIGLOLI		0.18KW/
		1-1.6A	1.2A		BONFIGLOLI		0.37KW
F11	K13 & K14	0.63-1A	0.78A	CHIP CONV.-2 MTR	LHPL / BONFIGLOLI		0.18KW/
		1-1.6A	1.2A		BONFIGLOLI		0.37KW
F12	K15 & K16	2.5-4A	2.76A	INDEX TABLE MTR	PRAGATI		1.6 KVA
F13	K17 & K18	0.63-1A	0.78A	SCREW CONV.MTR	LHPL / BONFIGLOLI		0.18KW/
		1-1.6A	1.2A		BONFIGLOLI		0.37KW
F14	-	0.01-0.16A		24VA TFR INPUT	-	-	-
F21	K23	0.24-0.40A		OIL SKIMMER OPTION			
F28	K26	2.5-4A		FUME EXTRACTOR			
F27	K25	1-1.6A	1.08A	COOLANT GUN MTR	RAJMANE	RK3-270	0.37KW
F33	K30	4-6.3A	5.4A	BAG FILTER MTR	RAJMANE	RKM-206	2.25KW

CONTACTORS	TYPE	MAKE	QTY
K	CA4-9C / 12C 24V DC	SPRECHER SHUH / A.B	8 NOS
PSM CONTACTOR	CA3-30C-01	SPRECHER SHUH	1
	CA3-43-N-11	SPRECHER SHUH	1
	CA3-72-N-11	SPRECHER SHUH	
ADD ON BLOCK	CS4-P	AB / SPRECHER SHUH	
EMG RELEASE	CS4C-40E	SPRECHER SHUH / AB 1 NO	1

		YES/NO	REMARKS
	ROTARY TABLE\ INDEX TABLE		
1	INTERFACING OF ROTARY TABLE (Complete)	Yes.	
2	INTERFACING OFINDEX TABLE	No	
3	ROTARY TABLE MAKE	✓	~~NIKKEN~~ \ UCAM ✓
5	INDEX TABLE STATIONS [8 \12 \24]	N.A	
6	INDEX TABLE CABLE LENGTH IN MTS	N.A	
7	ROTARY TABLE CABLE LENGTH IN MTS		
8	DRIVES FOR ROTARY TABLE [SVM BETA OR SVM]	—	
9	LOCATION OF MOUNTING OF JUNCTION BOX / CABINET [BOTTOM PLATE LH OR RH FROM FRONT]	R.H.	
10	SERVO TURRET OPTION	No	
	FIXTURE		
1	PNEUMATIC FIXTURE (NO.OF STATIONS)	No	
2	HYDRAULIC FIXTURE (NO.OF STATIONS)	No	
3	FIXTURE M .O. P OPTION	N.A	
4	COMP CLAMP OPTION WITH [1 / 2]PB IN MOP	N.A	
5	COMP DECLAMP OPTION WITH [1 / 2]PB IN MOP	N.A	
6	COMP CLAMP PR. SW OPTION	N.A	
7	COMP CLAMP AND DECLAMP PB(COMMON PB)	N.A	
8	EJECTOR UP/DOWN COMMON PB	N.A	
9	EJECTOR PRSW OPTION	N.A	

		YES/NO	REMARKS
10	COMP ORIENTATION PR. SW.OPTION		
11	COMP ORIENT AND DEORIENT PB(COMMON PB)		
12	COMP JACK LOCK OPTION WITH [1 / 2] PB IN MOP		
13	COMP JACK UNLOCK OPTION [1 / 2] PB IN MOP		
14	COMP JACK LOCK PR.SW OPTION		
15	JACK LOCK AND UNLOCK PB(COMMON PB)		
16	TAIL STOCK FWD OPTION WITH [1 / 2] PB IN MOP		
17	TAIL STOCK REV OPTION WITH [1 / 2] PB IN MOP		
18	TAIL STOCK FWD PRESSURE		
19	TAIL STOCK FWD AND REV PB(COMMON PB)		
20	COMP REST SENSE OPTION		
21	NO.OF REST SENSE SOLENOIDS [1 OR 2]		
22	REST SENSE PRESSURE SWITCHES [1 2 3 4]		LOW/HIGH
23	ANY OTHER SPECIFY		
24	FIXTURE SEQUENCE TO BE ATTACHED		
25	AUTO DOOR	YES/NO	

	OTHERS	YES/NO	REMARKS
1	AIR PRESSURE SWITCH	Yes	
2	PNEUMATIC COUNTER BALANCE	Yes	
3	MANUAL PALLET CHANGER	No	
4	HAND HELD MPG	No	
5	SEALED KEY BOARD \ MCP	✓	
6	Z-AXIS BRAKE (BUILTIN)	✓	
7	X & Y-AXIS BRAKE (BUILTIN / EXTNL)	No	
8	OIL SKIMMER OPTION	No.	
9	OIL COOLER OPTION	No	
10	CYCLE START SWITCH DHPB / SINGLE LEVER	No	
11	MIST EXTRACTOR OPTION	No	
12	SPINDLE LOAD METER (LM5)	No	
13	TOOL STATION DISPLAY UNIT (LM5)	Yes	
14	RENISHAW / MONTRONIX / M & H PROBE	No	
15	SEPARATE COOLANT GUN MTR	Yes.	
16	SERVO ARM OPTION	No	
17	SERVO MAGAZINE OPTION	No.	
18	NO. OF POCKETS IN MAGAZINE	20	
	APC		
1	LENGTH FOR APC PROXIMITY ON TABLE		M12 X 72 CONN. TYPE
2	TYPE OF LIMIT SWITCH FOR PALLET ON STATION		STRAIGHT / 90DEG
3	TYPE OF LIMIT SWITCH FOR DOOR UP AND DOWN		STRAIGHT / 90DEG

	CONFIRM FOLLOWING DATA		
1	KEEPRELAY AND TIMER DATA		
2	NC - PLC ARCHIVE W/O DD	✓	
3	NC - PLC ARCHIVE WITH DD	✓	
4	PLC PROGRAM	✓	
5	PART PROGRAM	✓	
6	HMI ARCHIVE	✓	
7	ALARM TEXT ARCHIVE	✓	
8	FIXTURE I/O'S, TB DETAILS ETC TO BE ATTACHED		
9	PUT BACKUP IN PC AND ENTER IN PENDING LIST		

	SPECIAL INFORMATIONS
1	802 SL
2	Rotary table (UCAM)
3	Std coolant.
4	
5	
6	

SIGNATURE:

NAME: Ghosh / Rajesh.

DATE:

ACE MANUFACTURING SYSTEMS LTD - BENGALURU

MACHINE CHECKLIST RECORD — QUALITY ASSURANCE

MACHINE NO: MA-467 MACHINE INCHARGE: Mr. Bhabu, Chandru.

SL.NO	MECHANICAL CHECK POINTS:
1	Counter Balance blkt pinning to be done.
2	Air reciver hose length to be reduced & proper routing.
3	Chip push motor remove the swivel elbow.
4	Coolant gun & Side flush hose to be changed.
5	Mechanical maintenace name plate to be changed.
6	Z axis limit Switch Cover to be fixed.
7	Spindle flange Cover to be fixed. & arround spl correct position.
8	Z axis refrence arrow blkt to be changed.
9	
10	
11	
12	
13	
14	
15	
16	
17	
18	
19	
20	

	ELECTRICAL CHECK POINTS:
1	Electrical maintenance name plate ~~to be~~ & m/c number plate
2	to be fixed.
3	
4	
5	
6	

QUALITY ASSURANCE ASSEMBLY MECHANICAL ELECTRICAL

Sl No	CHECK POINTS	AMS
I	AXIS	
1	Ensure plugs are fitted for L M Guide ways.	✓
2	Ensure effective wiping of saddle cover wipers.	T.Cover
3	Check the tightness of coupling and is set to 17/35Nm torque.	
4	Ensure the lub timer is set 12 min for HG and 20 min for LM Guide ways.	
5	Ensure leveling bolts are fixed with locknut.	✓
6	Check the Table tennon slot 14H8 / 18H8 (Tolerance 0 to +0.027 mm)	✓
7	Ensure the 3 axes Ball bar completion	✓
8	Check the all axes reference arrows are matched.	
9	The Machine/APC/Double Hand Emergency Push Button to be checked	✓
10	Ensure all screws marking neatly in Yellow paint marker	
11	Ensure cover fixed to Index table, rotary joint & tool breakage sensor	NA
12	Ensure sliding door free moment with out play and noise & nylon lock nut for rollers	✓
II	SPINDLE	
1	Check the belt tension by using belt tension meter.	D D
2	Ensure the proper supply of air to spindle at oriented condition.	✓
3	Ensure the proper supply of air to spindle Air Purge system	✓
4	Check the tightness of push stud.	✓
5	Ensure grub screws are fitted to spindle flange.	
6	Check spindle orientation position and note down the value. 4031 -	172-277
7	Check the spindle balancing	
III	ATC	
1	Ensure ATC / APC / Counter balance bracket pinning.	
2	Check oil in ATC gear box.	✓
3	Check the Tool not in spindle proximity message.	
4	Check the Tool Number display in panel board as same as in spindle	
IV	COOLANT	
1	Ensure no coolant leakage around the machine & lubrication oil drain pipe	
2	Ensure proper working of float switch.	
3	Ensure CTS flow switch setting	NA
4	Ensure suction strainer in CTS tank & pressure gauge working	NA
5	Ensure CTS pin locator hole has plugged	NA
6	Ensure gap b/w twin arm plunger CTS pin locator screw	NA
7	Ensure bag filter clogging.	NA
8	Ensure HP air blast after CTS switched off	NA
9	Ensure drum wash in conveyors	NA
10	Check conveyor overload message & Direction	NA
11	Ensure conveyor wheels are free from painting	✓
12	Check the function of M7, M30, M48, M52 & Coolant gun.	✓
V	APC	
1	Check pallet initialization in APC Machine	NA
2	Ensure effective supply of air jet to pallet rest pad cleaning.	
3	Ensure pallet change takes place smoothly without Heavy noise	

VI	HYD,PNEU & LUBRICATION	
1	Ensure air pressure switch is working properly.	
2	Ensure proper setting of oil in FRL unit. & Oil Level low message	
3	Check the FRL drain hose fixed.	
4	Ensure identification name plates are fixed in power pack hoses.	
5	Check the Hydraulic oil level low msg & Pressure gauge function	
6	Ensure no oil leakage in hydraulic valves & fittings	
7	Ensure lubrication pressure switch is set to 15 bar.	
	GENERAL	
1	Check all the points of MOM are attended	
2	Ensure coolant tank capacity engraved or marked in name plate	
3	Ensure Geometrical report, Practical test report, Spindle balancing report, Laser & Ballbar reports are attached in the history file	
4	Check Hydraulic circuit diagram pasted in power pack and specification is as per circuit diagram	

QUALITY ASSURANCE

DATE:

GEOMETRICAL TEST CHART

1. SPIRIT LEVEL READING (0.040 / mt):

Along X-Axis

Along Y-Axis

2. TABLE TOP READINGS (0.020 / 500 mm):

Along X-Axis

Along Y-Axis

3. T SLOT READING (0.020/500 mm):

PALLET - II

Along X-Axis

Along Y-Axis

4. COLUMN READINGS :

(0.020 / 300 mm)

FRONT

SIDE

5. MANDREL READINGS:

(0.020/ 300 mm)

Front

Side

RUNOUT:

a. TOP : (0.010mm)

1.

b. BOTTOM : (0.020mm)

1.

2.

3.

4.

6. TURN AROUND READING:

(0.020 / 300 mm Swing Dia)

7. SQUARENESS READING (X & Y):

(0.015 / 300 mm)

8. PALLET HEIGHT VARIATION:

(0.03 mm)

Pallet-1: _____

Pallet-2 : _____

9. PALLET REPEATABILITY:

(0.020mm)

	Pallet 1		Pallet 2	
	X	Y	X	Y
1.				
2.				
3.				
4.				
5.				

Quality Assurance	MACHINE CHECK LIST	AMS In Pursuit of Excellence

SL.No	PARTICULARS				
I	AXIS				
1	Check the marker pulse of all the axis : (Not applicable for β- Motors)	X	Y	Z	B
		—	—	—	—

2	Check the backlash of all the axis	X	Y	Z	B
	PA RAMETER : 1851 32450	0.002800	0.005500	0.003000	—
	PARAMETER : 1852 32452	1.000000	1.000000	1.000000	—
	PARAMETER 1850 (Grid Shift)	0	0	0	—

3	Check the Z - axis motor current during Tool de clamp:	
	a. WINNER / MCV 320 - 130 TO 150% - 840Feed rate in jog mode:	NA
	b. DTC - / 110 to 130 % Auto cycle	

4	Z - axis drop during power off :		
	a. + ve directions (Towards table) -	-230.000	-230.000
	b. - ve direction	-230.004	-230.004
	System :-	0.004	0.004
	Dial:-	0.002	0.001

II	AXIS				
1	% of Axis ~~current~~ Amps drawn: Amps				
	Axis	25% Rapid Rate	50% Rapid Rate	100% Rapiid Rate	600mm Feed Rate
	X	0.934	1.152	1.302	0.842
	Y	0.493	0.507	0.513	0.520
	Z- Down	0.487	0.490	0.493	0.480
	Z-UP	0.503	0.484	0.474	0.490
	B-Forw	1.400	1.500	2.2	
	B-Rev	1.300	1.600	2.25	

2	% of Axis holding ~~current~~ AMPS				
	Axis	100 % Rapid Rate		2000 mm feed rate	
		Motor End	Brake End	Motor End	Brake End
	X	-0.600	0.760	0.640	0.752
	Y	0.520	0.508	0.540	0.500
	Z	0.485	0.502	0.470	0.480

3	Over travel against Axis dead stopper in mm:				
	Sl.No	Axis	Motor End	Brake End	
	1	X	20.1	23.9	
	2	Y	21.9	10.0	
	3	Z	14.6	13.9	

III	SPINDLE			
1	Record the distance of push stud face from the spindle face when there is no tool			
	a. Tool ejection	Allowable	Auto	Manual
	BT - 40	0.7 / 1.0		
	BT - 30	0.5		
	b.Tool Clamping Force(Kgf)		580 Kgf	
2	Ensure that the gap between draw bar and ejection screw / Roller			
	a. When there is no tool in the spindle(1mm) :			
	b. When tool holder in the spindle (3mm) :			
3	Axial Play in the Spindle(mm):			
	a. Tool De- Clamping :	0 → +0.0017 →	Difference	
	b. 100% Load by Wood	0 → -0.0016 →	0.0002 mm	
4	a. % of ~~Current~~ Amps drawn at maximum RPM of the spindle :		0.843 Amps	
	b. Spindle noise at max RPM (db) [1 meter radious and 1.25 meter height]			
	b.1 ~~With~~ door close			
	b.2 ~~Without~~ door(77-80 db) open			
5	Spindle Stabilisation.		Room Temp: 30°C	
		Time	Spindle Temp	
	1	12:45.	30°C	
	2	1:15	38°C	
	3	1:50	41.5	
	Final Spindle Stabilisation Temperature :		41.5°C - 30°C = 11.5°C	
IV	A.T.C			
1	Check tool change noise (With tool holders)			
2	Ckeck the Spring lever compression (mm)		NA	
	a. Z -axis below to zero position.			
	b. After the magazine buts against the dead stopper:			
	c.Difference - BT-40 (2 to 3 mm) / BT-30 (1 to 2 mm)			
3	Tool picking position in mm		Actual	
			-13.100000	
4	Back Lash in the ATC disc gear (mm)			
V	A.P.C			
1	Presure setting for the pallet transfer cylinder in kg /cm2 :		NA	
2	Presure setting for the pallet Clamp - Declamp in kg /cm2 :			
VI	CYCLE TIME (IN SECONDS)			
1	Check the chip to chip time with Z axis only		3.18 Sec	
2	Check the tool to tool time			
3	Check the chip to chip time all axes		5.3 Sec	
4	Check the pallet changing cycle time		NA	

VII	TIME CONSTANT		X	Y	Z	B
1	Time constant linear - 1620	32300	4msec	4msec	4msec	
2	Time constant bell - 1621	32300	4msec	4msec	4msec	
3	Acceleration time in Sec					
4	Decceleration time in Sec :					

Quality Assurance	MACHINE HISTORY	AMS In Pursuit of Excellence

1	**STANDARD MACHINE :**				
			X - Axis	Y - Axis	Z- Axis
1.1	Ball Screw Pitch mm		12	12	12
1.2	Distance from Spindle face to table top with increased Day Light Area		mm	160	
2	**FLEXIBLE COUPLING :**		X - Axis	Y - Axis	Z- Axis
2.1	Model :				
2.2	Make :		Mayr		
3	**ATC DETAILS :**		4020V / 4024V / 4030V / 4020H / 4024H /		
3.1	ATC (Turret Type / Twin Arm)		Sl.No	7380	
3.2	Tool Capacity		Nos	6 / 12 / 16 / 20 / 24 / 30 / 40 /	
3.3	Side Cam Roller		Make	IKO	
			Model	CF-10 / CF-12	
			Make	Model	Sl.No
3.4	ATC Magazine Encoder		Pragati/Proximity	ATC12 / ATC16	
3.5	ATC Brake		EMCO		
3.6	ATC Arm Motor Brake		EMCO		
3.7	ATC Torque Motor		TM 50 / TM 80 / TM 100 / TM 125 / Servo ATC		
4	**AXIS MOTORS :**	4th axis B Axis	X -Axis	Y - Axis	Z-Axis
4.1	Model		1FK7083-5AF71-1FG2	1FK7083-5AF71-1FG2	1FK7083-5AF71-1FH0
4.2	Make :	FANUC / SIEMENS	Fanuc/ Siemens		
4.3	Sl.No		YFVD4777750I-001	YFVD4777750I 004	YFVD4777750I-02
4.4	Type				
5	**BRAKES :**				
5.1	Make		X Axis	Y Axis	Z Axis
			NA	NA	In Built
6	**HEAD STOCK DETAILS:**				
6.1	Spindle Details :			STD / CTS	
6.2	Spindle Motor			Specification	
	Make	Power 10.5/14.5	Fanuc/ Siemens	1PH7107-2QQ02-DKJ0	
	Power in Kw	10.5/13.0		YFVD4777800900I	
6.3	Spindle Nose Taper		BT	30 / 40 /50 / HSK	
6.4	Sl.No		08429		
6.5	Spindle Speed		RPM	80 - 8000	
6.6	Spindle Bearings Dia		&	50 / 55 / 60 / 70 / 80 /	
6.7	Spindle Drive			Direct Drive / Indirect Drive	
6.8	Pully Ratio		1:1 / 1:1.25 / 1:1.5 /1:1.75 / 1:2 /		
7	**BELT DETAILS : NOT APPLICABLE TO DIRECT DRIVE**				
			GT BELT	COUPLING	
7.1	Model			ROBA-DS, 10/951.221	
7.2	Make		Unitta / Power Grip	Mayr	

Quality Assurance

MACHINE HISTORY

8	**COOLANT SYSTEM DETAILS :**			
8.1	Tank Capacity (LTRS)	200 / 300 / 400		
	Rajmane Industries	Model		Type
8.2	Flood Coolant Pump / Flexinox	RKM 206 /	RV 160 / 270	Centrifugal Pump
8.3	Chip Pushing Coolant Pump	RKM 206 /	RV 160 / 270	
8.4	Base Flushing Coolant Pump	RKM 206 /	RV 160 / 270	
8.5	Coolant Gun & Drum Wash Pump	RKM 206 /	RV 160 / 270	
8.6	Oil Skimmer - Make / Model			
8.7	High Pressure Coolant Pump	Make	Grundfos/Wanner	
		Model		
8.8	Pressure Relief Valve	Make	Polyhydron	
		Model	DPRH-10T-100-11	
8.9	Rotary Joint	Make	Deublin / RIX	
		Model		
9	**FLOAT SWITCH :**			
9.1	Make	Filpro Elecdtronics		
9.2	Part No	L85247		
10	**CONVEYOR DETAILS :**			
10.1	Type of Conveyor :- Scraper along X or Y axis	Slat cum Scraper along X or Y axis		
10.2	Model			
10.3	Make	Universal Conveyor		
10.4	Sl.No			
11	**GEAR BOX DETAILS :**			
11.1	Model			
11.2	Make	Universal Conveyor		
12	**FILTER SPECIFICATIONS :**			
12.1	Model	8 /15 : Max Working Pressure 50 or 100 bar		
12.2	Make	Universal Preceision Tools		
12.3	Sl.No			
13	**COIL / SCREW CONVEYOR DETAILS :**			
13.1	Quantity			
13.2	Make	Universal Preceision Tools		
14	**HYDRAULIC SYSTEM :**			
14.1	Product No	Qc. No :		
14.2	Tank Capacity (LTRS)	32 / 40 / 63		
14.3	Maximum Working Pressure			
14.4	Make	Pascal / Hyd Tech / Rexorth / Yuken		
15	**PRESSURE SWITCHES :**			
15.1	Model	1PS 30-50 / 30-100 / 10 - 30 /		
15.1	Make	Polyhydron		
15.3	Quantity			

Quality Assurance

MACHINE HISTORY

16	**ACCUMULATOR :**	
16.12	Make	FOX / BOSCH
16.2	Set Oil Pressure	
16.3	Gas Pressure	
16.4	Volume in Lts	1.4 / 10

17	**HYDRAULIC CYLINDERS :**			
	Make	Model	Function	Qty
17.1	Island Hydarulic		Pallet F / R	2
17.2				
17.3				

18	**HYDRAULICS VALVES :**			
	Type	Make	Model	Qty
18.1	Directional Valve	Yuken	DSG-01-3C4-D24-N1-50	
18.2	Redusing Modular Valve	Yuken	MRP-01-B30	
18.3	Pilot operated check Valve	Yuken	MPW-01-2-30	
18.4	Throttle & Check Modular Valve	Yuken	MSW-01-X-30	
18.5				
18.11	Directional Valve	Rexroth	4WE6T 62/E G24 N9K4	
18.12	Redusing Modular Valve	Rexroth	ZDR6 DP 2-32/75YM	
18.13	Pilot operated check Valve	Rexroth	Z2S6-1-64	
18.14	Throttle & Check Modular Valve	Rexroth	Z2FS6-2-43/20QV	
18.15				

19	**HYDRAULIC PUMP :**	Make	Model
20	**MOTOR :**	Make	Model

21	**LUBRICATION UNIT DETAILS**	Make	Pressure Setting
		CENLUB / DROPCO	15 Kg/ cm2
	Motor	CENLUB / BATLIBOI	

22	**FRL UNIT :**			
	Make	Model	Operating Range	Grade of Oil
	Shavo	M4H-260-M68C	0-10 Kg/cm2	Servo Way Spin 12

23	**DRYER :**	Make	Specification
		Trident	VXD - 2

24	**PNEUMATIC VALVES :**			
	Application	Specs	Make	Qty
24.1	Spindle Air Jet	MFH 3-1/8	Festo	1
24.2	Spindle Air Purge	MFH 3-1/9	Festo	1
24.3	ATC Pocket up & Down	JMFH 5- 1/8	Festo	1
24.4	Manual Tool De Clamp	MFH 5-1/8	Festo	1
24.5	APC Auto Door	JMFH 5- 1/8	Festo	1
24.6	CTS Air Jet	MFH 3-1/8	Festo	1
24.7	Front Auto Door	JMFH 5-1/8	Festo	1
		MFH 3-1/8	Festo	1
24.8	Pallet Rest Pad Cleaning	MFH 3-1/8	Festo	1
24.9	Fixture	JMFH 5-1/8	Festo	

Regulator 1 — 10 to 16 — Festo — 1

Quality Assurance

MACHINE HISTORY

25	PRESSURE SWITCHES :				
25.1	Operating Range	2 to 15 Kg /cm2			
25.2	Make	SUCO			
25.3	Quantity	1 + 1			
26	PNEUMATIC CYLINDERS USED(BROUGHT OUT IF ANY) :				
	Function	Make	Model	Qty	
	Manual Tool De Clamp	Festo	DNU80-80-PPV DNU-100-60-PPV	1	
		Rotex	SCN D -100 / 60 SCN D 80 / 80	1	
	Front Auto Door	Festo			
	APC Auto Door	Festo		1	
27	TURRET : INDEXING TABLE (OPTIONAL)				
	Make	Model	Sl.No	No of Station	
	Pragathi / AMS			8 /12 /	
28	ROTARY TABLE	Make	Model	Sl.No	
		NIKKEN / UCAM	VRH-201- RH	M10-1996-860	
29	TAIL STOCK	Make	Model	Sl.No	
		UCAM	VTAIL-140-M	M10-1996-851	
30	LIMIT SWITCHES / READ SWITCHS				
	Make	Qty	Function		
30.1	Jai Balaji / Teknic Euncher	1	Front Auto Door / Front Door With CTS		
30.2	Jai Balaji / Teknic Euncher	2	APC Auto Door		
30.3	Jai Balaji / Teknic Euncher	2	Pallet in Station		
31	PNEUMATIC COUNTER BALANCING SMC / Rotex / Festo		Specification		
31.1	Cylinder Dia : 80 / 100 / 125		ITMB1A0-X3M006-600		
31.2	Filter		1-AFM40-04B		
31.3	Pressure Regulator		1R 3020 - 04 BGI		
31.4	Process Valve		VNA 20 1A - 15A		
32	Component Seat check				
	Make	Qty	Model		
	SMC / Norgen				
33	Air reservoir				
	Capacity	30 / 40 / 60			
34	SPECIAL INFORMATIONS / REMARKS				
1	2 axis Tool change limit switch provided.				
2	4th axis is not our Scope. only for data				
3					
4					
5					

Rotary Table Mock Vacuum
DPRH-201-RH
SNo MTO 1998-R80

Tool stock - Ucam
VTAIL-140 M
No MTO 1996-851

ACE MANUFACTURING SYSTEMS LTD., AMS

GEOMETRICAL TEST CHART

PRODUCT : V M C

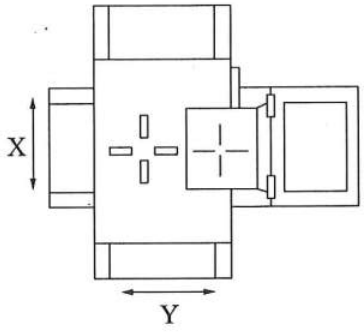
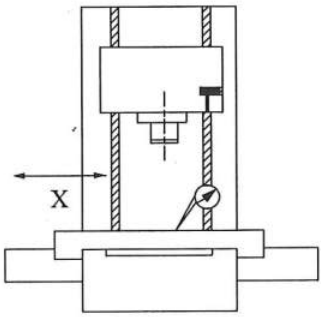
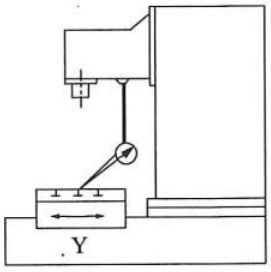
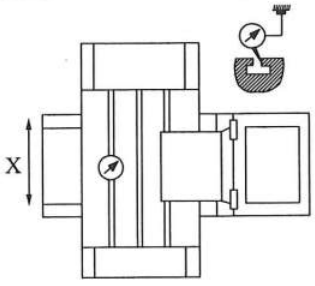
MODEL NO. : MCV-400

MACHINE NO. M11-467

CUSTOMER : ACUTECH

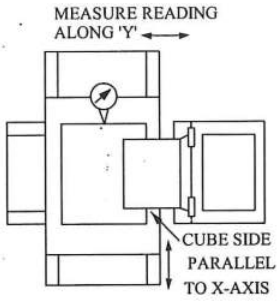
DATE : 21-02-09	CHD. CUST. :	AMD. DT. :	DOC. NO. D11-001-01-003
BY :	CHD. AMS 21/02	SHEET : 01 OF 05	PAGE NO. F3

AMS	TEST CHART FOR VERTICAL MACHINING CENTER GEOMETRICAL ACCURACY TESTS	MCV-400 M11=467

SL. NO.	Description of the test	Figure	All measurements in mm Tolerance Permissible	Actual
G1	Levelling of Machine X & Y axes at their mid stroke		0.04/m along X & Y axes.	Level along X-axis : 0.01 Level along Y-axis : 0.01
G2	Parallelism of table surface w.r.t. its Guideways or X-axis movement		0.025/600	0.006 0.006 0.004
G3	Parallelism of table surface w.r.t. Y-axis Guideways or Y-axis movement		0.02/415	0.008 0.005 0.005
G4	Parallelism of reference T-slot to X-axis movement		0.025/600	0.010

DATE : 21-02-09	CHD. CUST. :	AMD. DT. :	DOC. NO. D11-001-01-03
BY :	CHD. AMS :	SHEET : 02 OF 05	PAGE NO. F3

AMS	TEST CHART FOR VERTICAL MACHINING CENTER GEOMETRICAL ACCURACY TESTS	MCV-400 M11-467

SL. NO.	Description of the test	Figure	All measurements in mm Tolerance Permissible	Actual
G13	Squareness of Y-axis movement w.r.t. X-axis movement		0.02/460	0.007

DATE : 21-02-09	CHD. CUST. :	AMD. DT. :	DOC. NO. D11-001-01-03
BY :	CHD. AMS :	SHEET : 05 OF 05	PAGE NO. F3

ACE MANUFACTURING SYSTEMS LTD.,

GEOMETRICAL TEST CHART

PRODUCT : VERTICAL M/C CENTER

MODEL NO : MCV-400

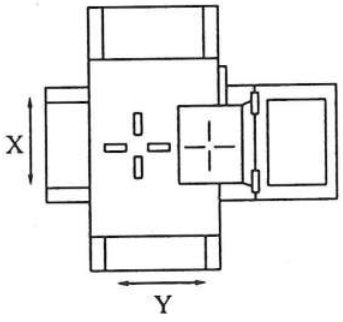
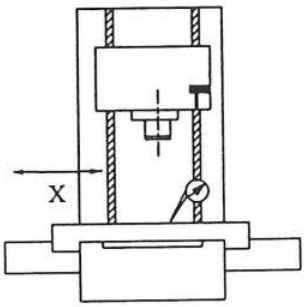
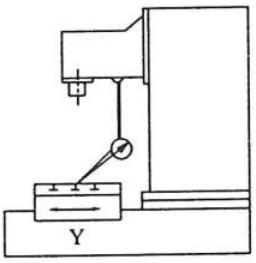
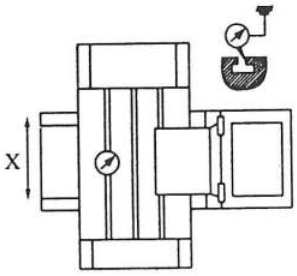
MACHINE NO: M11-467

CUSTOMER : ACUTECH

DATE :	QA :	DOC No. :	SHEET :
CHECKED BY AMS :	CHECKED BY CUSTOMER :	PAGE No. :	F3 :

Corrected test chart attached,

AMS	TEST CHART FOR VERTICAL MACHINING CENTER GEOMETRICAL ACCURACY TESTS	MCV400 M11

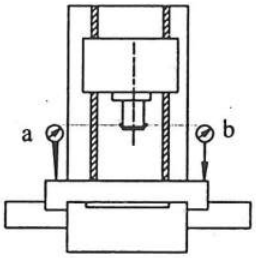
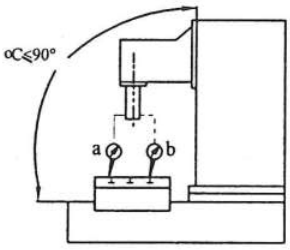
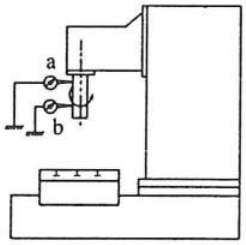
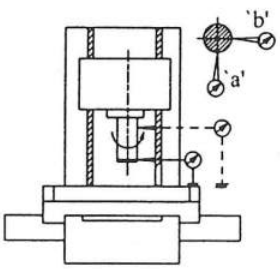
SL. NO.	Description of the test	Figure	All measurements in mm Tolerance: Permissible	Actual
G1	Levelling of Machine X & Y axes at their mid stroke		0.04/m along X & Y axes. 0.010 0.005	Level along X-axis : 0 Level along Y-axis : 0
G2	Parallelism of table surface w.r.t. its Guideways or X-axis movement		0.025/600 0 +8 +6 0 +4 +6 0 +2 +2	
G3	Parallelism of table surface w.r.t. Y-axis Guideways or Y-axis movement		0.02/415 0 0 0 +5 +2 -5 +3 -4 -8	
G4	Parallelism of reference T-slot to X-axis movement		0.025/600 0.006	

DATE :	CHD. CUST. :	AMD. DT. :	DOC. NO. D11-001-01-03	
BY :	CHD. AMS :	SHEET : 01 OF 04	PAGE NO.	F3

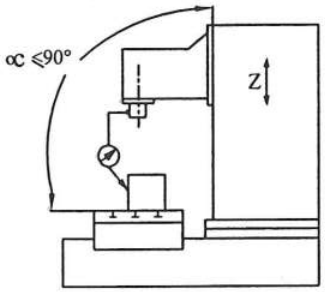
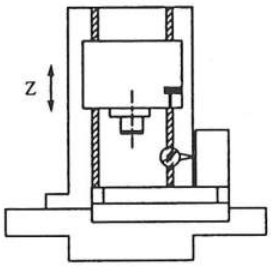
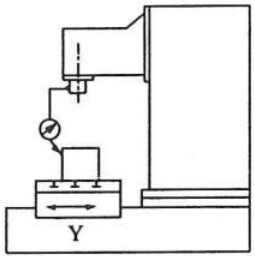
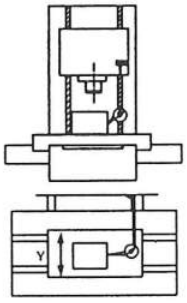
AMS

TEST CHART FOR VERTICAL MACHINING CENTER GEOMETRICAL ACCURACY TESTS

MCV400
M11

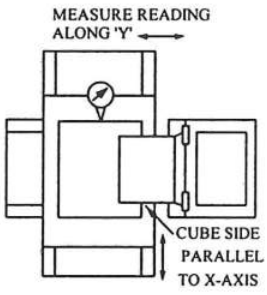
SL. NO.	Description of the test	Figure	All measurements in mm Tolerance Permissible	Actual
G5	Squareness of the Spindle axis to the table surface in the XZ plane. X & Y axes at their mid stroke		0.020 for 300 swing diameter	0.010
G6	Squareness of the Spindle axis to the table surface in the YZ plane. X & Y axes at their mid stroke		0.02 for 300 swing diameter ∝≤90°	0.009
G7	Run out of spindle taper a) Near the Spindle nose b) At a distance of 300 from the spindle nose		a) 0.01 near the spindle nose b) 0.02 at a dist. of 300mm.	a. 0.003 b. ~~0.010~~ 0.005
G8	Parallelism of spindle Axis & Z-axis in a) XZ plane b) YZ plane to be checked at mean point of runout		a) 0.020/300 b) 0.020/300	a. 0.0012 b. 0.008

AMS	TEST CHART FOR VERTICAL MACHINING CENTER GEOMETRICAL ACCURACY TESTS	*MCV400* *M11*

SL. NO.	Description of the test	Figure	All measurements in mm Tolerance Permissible	Actual
G9	Squareness between table surface & the spindle head along the Z-axis in XZ plane		0.03/460 -0.002	
G10	Squareness between table surface & the spindle head along the Z-axis in YZ plane		0.03/460 +0.002	
G11	Straightness of X-axis movement in XZ plane		0.01/500 0.001	
G12	Straightness of Y-axis movement in YZ plane		0.01/415 0.006	

DATE :	CHD. CUST. :	AMD. DT. :	DOC. NO. D11-001-01-03	
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AMS	TEST CHART FOR VERTICAL MACHINING CENTER GEOMETRICAL ACCURACY TESTS	*MCV400* *M11*

SL. NO.	Description of the test	Figure	All measurements in mm Tolerance Permissible	Actual
G13	Squareness of Y-axis movement w.r.t. X-axis movement		0.02/460	

DATE :	CHD. CUST. :	AMD. DT. :	DOC. NO. D11-001-01-03	
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ACE MANUFACTURING SYSTEMS LTD., AMS

DESPATCH TIME CHECKING

GEOMETRICAL TEST CHART

PRODUCT : V M C

MODEL NO. : MCV-400

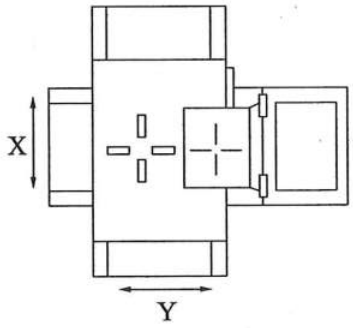
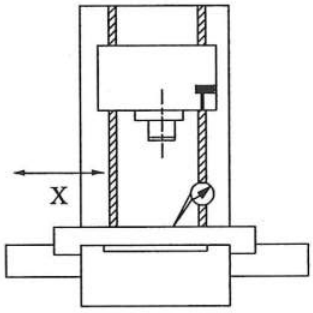
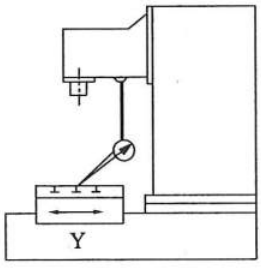
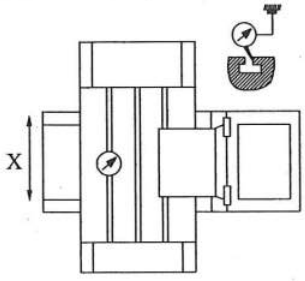
MACHINE NO. M11-467

CUSTOMER : ACUTECH

NOTE :- Column Reading is more.
That is Correction Colum Reading. Test Chart.

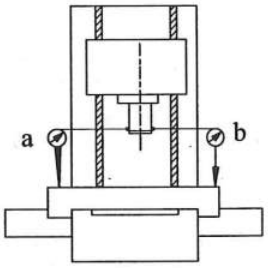
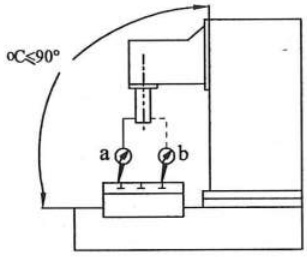
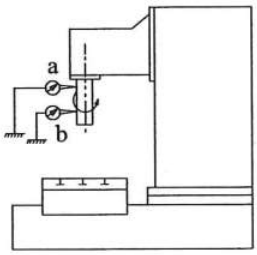
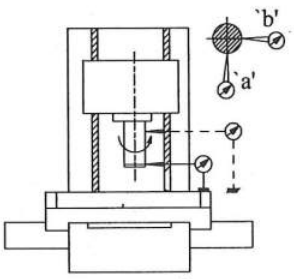
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BY :	CHD. AMS :	SHEET : 01 OF 05	PAGE NO. F3

AMS	TEST CHART FOR VERTICAL MACHINING CENTER GEOMETRICAL ACCURACY TESTS	*MCV-400* *M11-467*

SL. NO.	Description of the test	Figure	All measurements in mm Tolerance Permissible	Actual
G1	Levelling of Machine X & Y axes at their mid stroke		0.04/m along X & Y axes.	Level along X-axis : 0.01 Level along Y-axis : 0.01
G2	Parallelism of table surface w.r.t. its Guideways or X-axis movement		0.025/600	0.006 0.006 0.004
G3	Parallelism of table surface w.r.t. Y-axis Guideways or Y-axis movement		0.02/415	0.008 0.005 0.005
G4	Parallelism of reference T-slot to X-axis movement		0.025/600	0.010

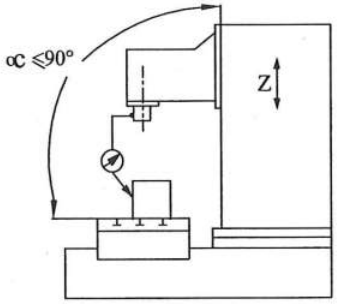
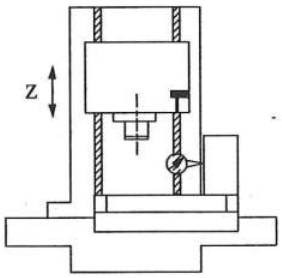
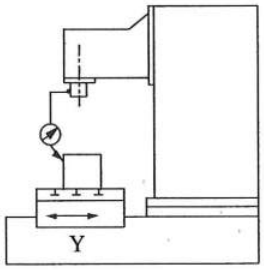
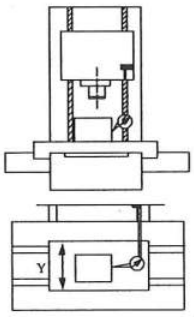
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BY :	CHD. AMS :	SHEET : 02 OF 05	PAGE NO. F3

AMS	TEST CHART FOR VERTICAL MACHINING CENTER GEOMETRICAL ACCURACY TESTS	MCV-400 M11- 467

SL. NO.	Description of the test	Figure	All measurements in mm Tolerance Permissible	Actual
G5	Squareness of the Spindle axis to the table surface in the XZ plane. X & Y axes at their mid stroke		0.020 for 300 swing diameter	0.006
G6	Squareness of the Spindle axis to the table surface in the YZ plane. X & Y axes at their mid stroke		0.02 for 300 swing diameter oC≤90°	0.008
G7	Run out of spindle taper a) Near the Spindle nose b) At a distance of 300 from the spindle nose		a) 0.01 near the spindle nose b) 0.02 at a dist. of 300mm.	a. 0.002 b. 0.004
G8	Parallelism of spindle Axis & Z-axis in a) XZ plane b) YZ plane to be checked at mean point of runout		a) 0.020/300 b) 0.020/300	a. 0.002 b. 0.003

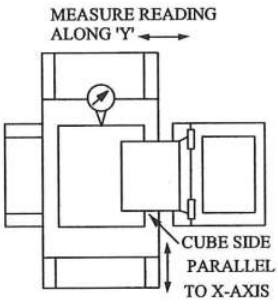
DATE : 28-05-09	CHD. CUST. :	AMD. DT. :	DOC. NO. D11-001-01-03
BY :	CHD. AMS :	SHEET : 03 OF 05	PAGE NO. F3

AMS	TEST CHART FOR VERTICAL MACHINING CENTER GEOMETRICAL ACCURACY TESTS	MCV-400 M11-467

SL. NO.	Description of the test	Figure	All measurements in mm Tolerance Permissible	Actual
G9	Squareness between table surface & the spindle head along the Z-axis in XZ plane		0.03/460	0.005 [Corrected paramet
G10	Squareness between table surface & the spindle head along the Z-axis in YZ plane		0.03/460	0.008
G11	Straightness of X-axis movement in XZ plane		0.01/500	0.002
G12	Straightness of Y-axis movement in YZ plane		0.01/415	0.002

DATE : 28-05-09	CHD. CUST. :	AMD. DT. :	DOC. NO. D11-001-01-03
BY :	CHD. AMS :	SHEET : 04 OF 05	PAGE NO. F3

AMS	TEST CHART FOR VERTICAL MACHINING CENTER GEOMETRICAL ACCURACY TESTS	MCV-400 M11 – 467

SL. NO.	Description of the test	Figure	All measurements in mm Tolerance Permissible	Actual
G13	Squareness of Y-axis movement w.r.t. X-axis movement		0.02/460	0.007

DATE : 28-05-09	CHD. CUST. :	AMD. DT. :	DOC. NO. D11-001-01-03	
BY :	CHD. AMS :	SHEET : 05 OF 05	PAGE NO.	F3

RENISHAW CALIBRATION INTERFEROMETER SYSTEM

ENVIRONMENT DATA PRINT-OUT

Mode : Linear displacement

File: M11-467-X1.RTL

Machine:MCV-400.
Date:11:39 Feb 16 2009
Axis:X.

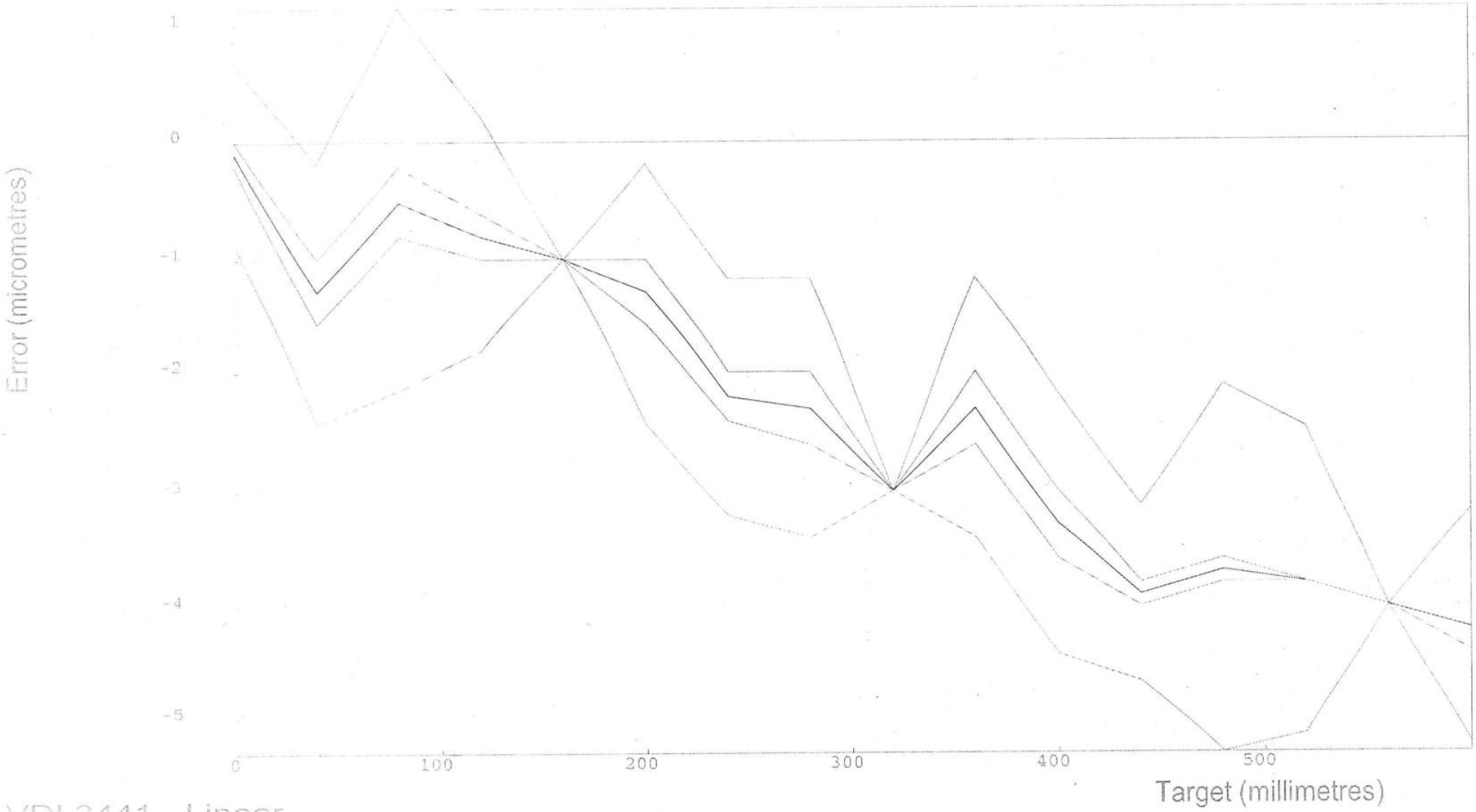
Serial No:M11-467.
By:QAS/SA.
Location:AMS.

TITLE:M/s.ACUTECH.

Data	Start	End	Units
Time :	11:39 Feb 16 2009	11:48 Feb 16 2009	
Air temperature :	26.13	26.24	deg C
Air pressure :	911.20	911.20	mbar
Rel. humidity :	40.31	39.81	%
Material Temperature 1 :	23.05	23.18	deg C
Material Temperature 2 :	--NC--	--NC--	deg C
Material Temperature 3 :	--NC--	--NC--	deg C
Exp. Coeff :	11.70	11.70	ppm/deg C
Environment factor :	0.31640841	0.31640798	

VDI 3441

M/s.ACUTECH.



VDI 3441 - Linear

Machine:MCV-400.	Axis:X.	U max : 0.600
Serial No:M11-467.	Location:AMS.	P : 6.434
Date:11:39 Feb 16 2009	U mean : 0.338	Ps max : 2.985
By:QAS/SA.	Ps mean : 1.511	Pa : 4.100

RENISHAW CALIBRATION INTERFEROMETER SYSTEM
ALL DATA

Machine:MCV-400. Serial No:M11-467.
Date:11:39 Feb 16 2009 By:QAS/SA.
Axis:X. Location:AMS.
TITLE:M/s.ACUTECH. Filename: M11-467-X1.RTL

Units of data values are micrometres

Target 1 (+) (0.0000 millimetres)
1) 0.0 2) 0.0 3) 0.0 4) 0.0
5) 0.0
Target 1 (-)
1) 0.0 2) 0.0 3) 0.0 4) 0.0
5) -1.0

Target 2 (+) (39.9990 millimetres)
1) -1.0 2) -1.0 3) -1.0 4) -1.0
5) -1.0
Target 2 (-)
1) -1.0 2) -1.0 3) -2.0 4) -2.0
5) -2.0

Target 3 (+) (79.9980 millimetres)
1) 0.0 2) 0.0 3) 0.0 4) 0.0
5) -1.0
Target 3 (-)
1) 0.0 2) -1.0 3) -1.0 4) -1.0
5) -1.0

Target 4 (+) (119.9970 millimetres)
1) 0.0 2) 0.0 3) -1.0 4) -1.0
5) -1.0
Target 4 (-)
1) -1.0 2) -1.0 3) -1.0 4) -1.0
5) -1.0

Target 5 (+) (159.9960 millimetres)
1) -1.0 2) -1.0 3) -1.0 4) -1.0
5) -1.0
Target 5 (-)
1) -1.0 2) -1.0 3) -1.0 4) -1.0
5) -1.0

Target 6 (+) (199.9950 millimetres)
1) -1.0 2) -1.0 3) -1.0 4) -1.0
5) -1.0
Target 6 (-)
1) -1.0 2) -1.0 3) -2.0 4) -2.0
5) -2.0

Target 7 (+) (239.9940 millimetres)
1) -2.0 2) -2.0 3) -2.0 4) -2.0
5) -2.0
Target 7 (-)
1) -2.0 2) -2.0 3) -2.0 4) -3.0
5) -3.0

Target 8 (+) (279.9930 millimetres)
1) -2.0 2) -2.0 3) -2.0 4) -2.0
5) -2.0

Target 8 (-)
1) -2.0 2) -2.0 3) -3.0 4) -3.0
5) -3.0

Target 9 (+) (319.9920 millimetres)
1) -3.0 2) -3.0 3) -3.0 4) -3.0
5) -3.0
Target 9 (-)
1) -3.0 2) -3.0 3) -3.0 4) -3.0
5) -3.0

Target 10 (+) (359.9910 millimetres)
1) -2.0 2) -2.0 3) -2.0 4) -2.0
5) -2.0
Target 10 (-)
1) -2.0 2) -2.0 3) -3.0 4) -3.0
5) -3.0

Target 11 (+) (399.9900 millimetres)
1) -3.0 2) -3.0 3) -3.0 4) -3.0
5) -3.0
Target 11 (-)
1) -3.0 2) -3.0 3) -4.0 4) -4.0
5) -4.0

Target 12 (+) (439.9890 millimetres)
1) -3.0 2) -4.0 3) -4.0 4) -4.0
5) -4.0
Target 12 (-)
1) -4.0 2) -4.0 3) -4.0 4) -4.0
5) -4.0

Target 13 (+) (479.9880 millimetres)
1) -3.0 2) -3.0 3) -4.0 4) -4.0
5) -4.0
Target 13 (-)
1) -4.0 2) -4.0 3) -3.0 4) -4.0
5) -4.0

Target 14 (+) (519.9870 millimetres)
1) -3.0 2) -4.0 3) -4.0 4) -4.0
5) -4.0
Target 14 (-)
1) -4.0 2) -4.0 3) -3.0 4) -4.0
5) -4.0

Target 15 (+) (559.9860 millimetres)
1) -4.0 2) -4.0 3) -4.0 4) -4.0
5) -4.0
Target 15 (-)
1) -4.0 2) -4.0 3) -4.0 4) -4.0
5) -4.0

Target 16 (+) (599.9850 millimetres)
1) -4.0 2) -4.0 3) -4.0 4) -4.0
5) -4.0
Target 16 (-)
1) -4.0 2) -4.0 3) -4.0 4) -5.0
5) -5.0

RENISHAW CALIBRATION INTERFEROMETER SYSTEM

ENVIRONMENT DATA PRINT-OUT

Mode : Linear displacement

File: M11-467-Y1.RTL

Machine:MCV-400.
Date:12:16 Feb 16 2009
Axis:Y.

Serial No:M11-467.
By:QAS/SA.
Location:AMS.

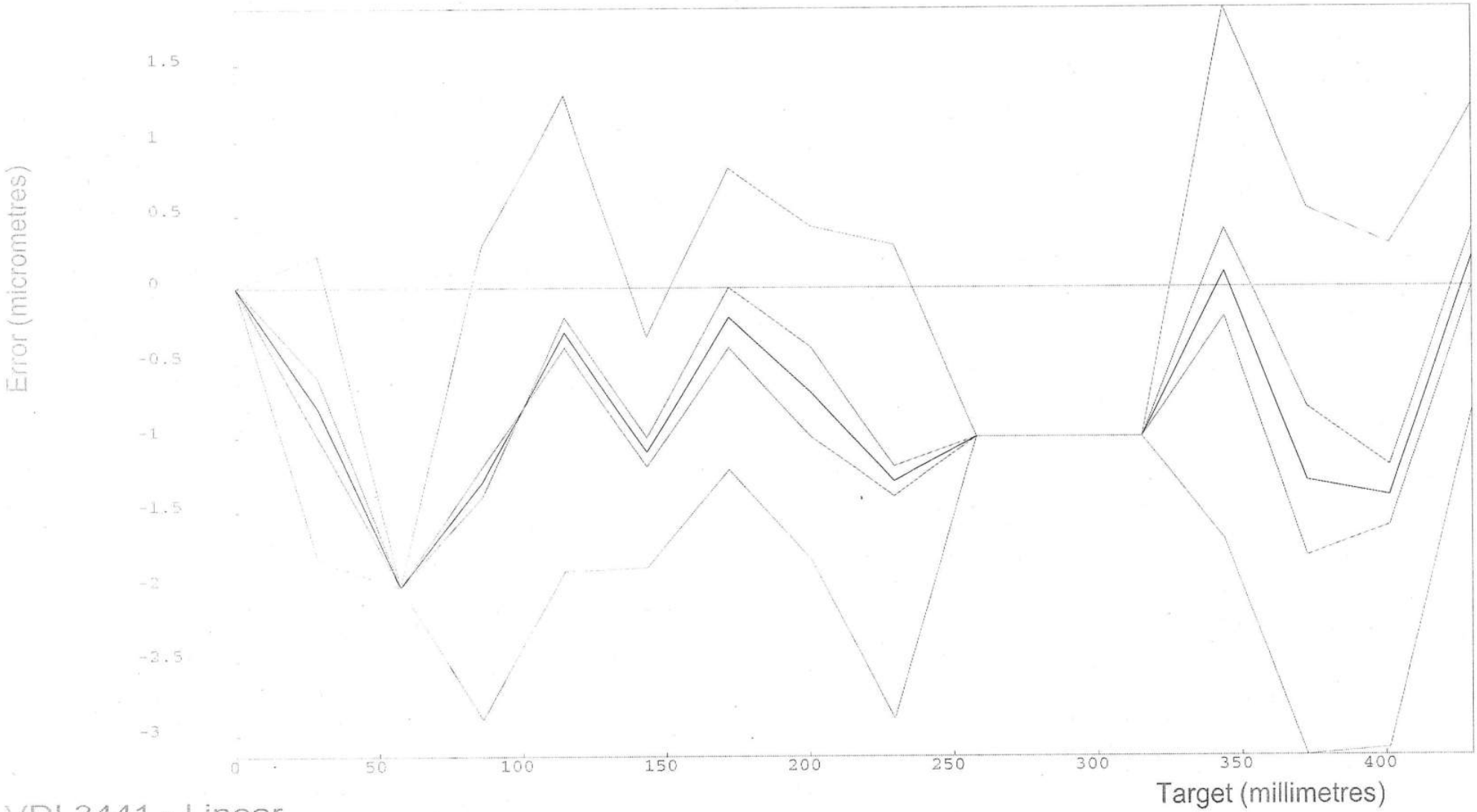
TITLE:M/s.ACUTECH.

Data	Start	End	Units
Time :	12:16 Feb 16 2009	12:26 Feb 16 2009	
Air temperature :	26.62	26.78	deg C
Air pressure :	910.60	910.60	mbar
Rel. humidity :	39.57	38.06	%
Material Temperature 1 :	23.50	23.66	deg C
Material Temperature 2 :	--NC--	--NC--	deg C
Material Temperature 3 :	--NC--	--NC--	deg C
Exp. Coeff :	11.70	11.70	ppm/deg C
Environment factor :	0.31640694	0.31640637	



VDI 3441

M/s.ACUTECH.



VDI 3441 - Linear

Machine:MCV-400.	Axis:Y.	U max : 1.000
Serial No:M11-467.	Location:AMS.	P : 5.034
Date:12:16 Feb 16 2009	U mean : 0.287	Ps max : 2.985
By:QAS/SA.	Ps mean : 1.595	Pa : 2.200

RENISHAW CALIBRATION INTERFEROMETER SYSTEM
ALL DATA

Machine:MCV-400. Serial No:M11-467.
Date:12:16 Feb 16 2009 By:QAS/SA.
Axis:Y. Location:AMS.
TITLE:M/s.ACUTECH. Filename: M11-467-Y1.RTL

Units of data values are micrometres

Target 1 (+) (0.0000 millimetres)
1) 0.0 2) 0.0 3) 0.0 4) 0.0
5) 0.0
Target 1 (-)
1) 0.0 2) 0.0 3) 0.0 4) 0.0
5) 0.0

Target 2 (+) (28.6670 millimetres)
1) -1.0 2) -1.0 3) -1.0 4) -1.0
5) -1.0
Target 2 (-)
1) -1.0 2) -1.0 3) -1.0 4) 0.0
5) 0.0

Target 3 (+) (57.3340 millimetres)
1) -2.0 2) -2.0 3) -2.0 4) -2.0
5) -2.0
Target 3 (-)
1) -2.0 2) -2.0 3) -2.0 4) -2.0
5) -2.0

Target 4 (+) (86.0010 millimetres)
1) -2.0 2) -2.0 3) -1.0 4) -1.0
5) -1.0
Target 4 (-)
1) -2.0 2) -1.0 3) -1.0 4) -1.0
5) -1.0

Target 5 (+) (114.6680 millimetres)
1) -1.0 2) 0.0 3) 0.0 4) 0.0
5) 0.0
Target 5 (-)
1) -1.0 2) -1.0 3) 0.0 4) 0.0
5) 0.0

Target 6 (+) (143.3350 millimetres)
1) -1.0 2) -1.0 3) -1.0 4) -1.0
5) -1.0
Target 6 (-)
1) -1.0 2) -2.0 3) -1.0 4) -1.0
5) -1.0

Target 7 (+) (172.0020 millimetres)
1) 0.0 2) 0.0 3) 0.0 4) 0.0
5) 0.0
Target 7 (-)
1) -1.0 2) -1.0 3) 0.0 4) 0.0
5) 0.0

Target 8 (+) (200.6690 millimetres)
1) -1.0 2) -1.0 3) 0.0 4) 0.0
5) 0.0

Target 8 (-)
1) -1.0 2) -1.0 3) -1.0 4) -1.0
5) -1.0

Target 9 (+) (229.3360 millimetres)
1) -2.0 2) -1.0 3) -1.0 4) -1.0
5) -1.0
Target 9 (-)
1) -2.0 2) -2.0 3) -1.0 4) -1.0
5) -1.0

Target 10 (+) (258.0030 millimetres)
1) -1.0 2) -1.0 3) -1.0 4) -1.0
5) -1.0
Target 10 (-)
1) -1.0 2) -1.0 3) -1.0 4) -1.0
5) -1.0

Target 11 (+) (286.6700 millimetres)
1) -1.0 2) -1.0 3) -1.0 4) -1.0
5) -1.0
Target 11 (-)
1) -1.0 2) -1.0 3) -1.0 4) -1.0
5) -1.0

Target 12 (+) (315.3370 millimetres)
1) -1.0 2) -1.0 3) -1.0 4) -1.0
5) -1.0
Target 12 (-)
1) -1.0 2) -1.0 3) -1.0 4) -1.0
5) -1.0

Target 13 (+) (344.0040 millimetres)
1) 0.0 2) 0.0 3) 0.0 4) 1.0
5) 1.0
Target 13 (-)
1) -1.0 2) 0.0 3) 0.0 4) 0.0
5) 0.0

Target 14 (+) (372.6710 millimetres)
1) -1.0 2) -1.0 3) -1.0 4) -1.0
5) 0.0
Target 14 (-)
1) -2.0 2) -2.0 3) -2.0 4) -2.0
5) -1.0

Target 15 (+) (401.3380 millimetres)
1) -2.0 2) -1.0 3) -1.0 4) -1.0
5) -1.0
Target 15 (-)
1) -2.0 2) -2.0 3) -2.0 4) -1.0
5) -1.0

Target 16 (+) (430.0050 millimetres)
1) 0.0 2) 0.0 3) 0.0 4) 1.0
5) 1.0
Target 16 (-)
1) 0.0 2) 0.0 3) 0.0 4) 0.0
5) 0.0

Ballbar diagnostics table
M11-467-XY-A2

RENISHAW

Operator: Administrator
Date: 2009-Feb-16 12:57:56

Machine: Quick check
Instrument: QC10 ballbar

Error	Magnitude		Independent circularity		Ranking
Backlash X	▸-0.3	◂0.0 µm	0.3 µm	(1%)	(12)
Backlash Y	▴0.5	▾0.4 µm	0.5 µm	(2%)	(10)
Reversal spikes X	▸0.5	◂-1.0 µm	1.6 µm	(7%)	(4)
Reversal spikes Y	▴-4.4	▾-5.4 µm	5.4 µm	(25%)	(1)
Lateral play X	▸-0.5	◂0.1 µm	0.3 µm	(1%)	(11)
Lateral play Y	▴-1.2	▾1.0 µm	0.6 µm	(3%)	(8)
Cyclic error X	↑0.6	↓0.5 µm	0.6 µm	(3%)	(7)
Cyclic error Y	↑1.0	↓1.1 µm	1.1 µm	(5%)	(6)
Servo mismatch	0.01 ms		0.2 µm	(1%)	(13)
Squareness	-19.1 µm/m		2.9 µm	(13%)	(2)
Straightness X	1.0 µm		0.5 µm	(2%)	(9)
Straightness Y	2.3 µm		1.2 µm	(5%)	(5)
Scaling mismatch	-4.2 µm		2.1 µm	(10%)	(3)
Cyclic pitch X	12.0000 mm				
Cyclic pitch Y	12.0000 mm				
Scaling error X	-22.3 ppm				
Scaling error Y	-8.4 ppm				
Calculated feedrate	998.6 mm/min				
Centre offset X	-0.7 µm				
Centre offset Y	-3.1 µm				
Positional tolerance	19.4 µm				
Best fit radius	149.9977 mm				
Circularity	8.5 µm				

Test Parameters

Radius	150.0000mm
Feedrate	1000.0mm/min
Start/End/Overshoot	0°/0°/180°
Run sequence	CW CCW
Test position	
Sample rate	27.778Hz

ISO 230-4:1996(E) Circular Deviation
M11-467-XY-A2

Operator: Administrator
Date: 2009-Feb-16 12:57:56

RENISHAW

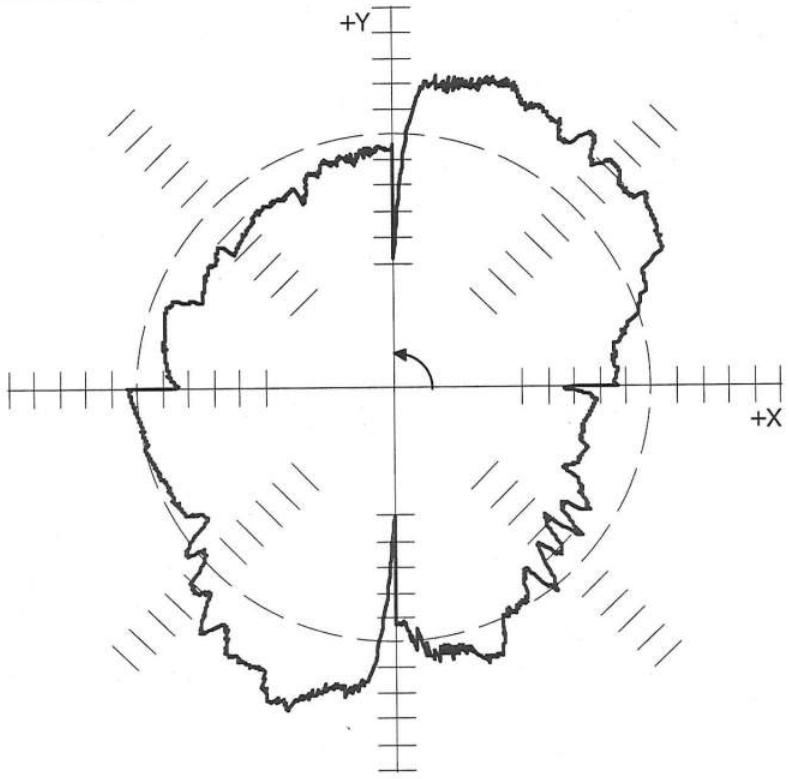
Machine: Quick check
Instrument: QC10 ballbar

Circular Deviation (CW)

Value	8.3µm

Test Parameters

Radius	150.0000mm
Sample rate	27.778Hz
Feedrate	1000.0mm/min
Run direction	CW
Plane under test	XY
Test position	
Start angle	0°
End angle	0°
Overshoot angle	180°



Run 1

1.0 µm/div

ISO 230-4:1996(E) Circular Deviation
M11-467-XY-A2

Operator: Administrator
Date: 2009-Feb-16 12:57:56

RENISHAW

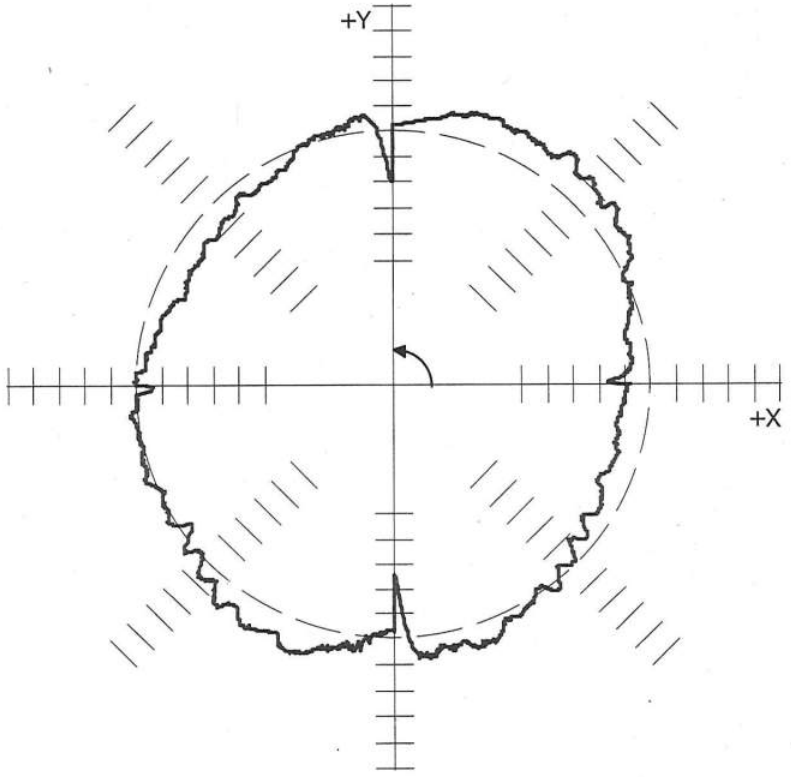
Machine: Quick check
Instrument: QC10 ballbar

Circular Deviation (CCW)

Value	8.1µm

Test Parameters

Radius	150.0000mm
Sample rate	27.778Hz
Feedrate	1000.0mm/min
Run direction	CCW
Plane under test	XY
Test position	
Start angle	0°
End angle	0°
Overshoot angle	180°



Run 2

2.0 µm/div